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GitHub Link

Includes dataset etc

An investigation into how a Sobriety Tracing app helps people to stay sober and what keeps people motivated to hitting their goals.

Outline Description / Research Question

This project will be the design and development of a sobriety app that the goal is to help individuals maintain sobriety and develop healthier habits. Individuals attempting to stop or reduce harmful behaviour and may experience challenges like cravings, low motivations, stress and may struggle to maintain consistency.

The app will let users track how long they have maintained sobriety, record their personal reasons for quitting, review their progress and confirm their commitment through daily check-ins. In addition to developing the app, the project will investigate the factors that influence sobriety, including personal backgrounds, motivations, challenges and support systems. Relevant existing public datasets will be used to explore and inform the design of my project.

Research Questions

I want to learn how a digital sobriety support app can use progress tracking, daily reflection and motivational features to help individuals maintain sobriety?

Supporting Questions

- What Factors influence an individual's motivation to pursue sobriety?
- What challenges do individuals face when trying to maintain sobriety?
- Which digital feature may help users remain motived and engaged?
- How can research and data analysis inform the design of a user-friendly sobriety application?

Proposed Project Artefact / Project Deliverable(s):

The main function will be a prototype of sobriety support application which will provide users with a simple way to track their progress, reflect on their motivations and engage with support features through their journey.

Features

User Registration and Login

Users will be able to create an account and securely access their personal information and progress.

Homepage and Navigation

A homepage will provide an overview of the user's progress with a navigation menu allow access to the main features.

Sobriety Tracker

Users will be able to select the habit or substance they wish to stop using, record their start date and view a counter showing their progress.

Progress Summary

The application will be able to record and edit their reasons for pursuing sobriety. These reasons can be revisited when they need encouragement.

Daily Check In

A dedicated “Today” page will allow users to confirm their daily commitment, reflect on their experiences and record optional notes about their progress.

Motivation / Reflection Page

This page will provide supporting messages, personal reminders and opportunities to reflect on achievements and challenges.

Settings

Users will be able to manage their preference, edit their information and reset or restart their tracking when needed.

Proposed Development Tools / Platforms

- Frontend: Either HTML, CSS and JavaScript or PowerApps default creation tools. This will help me to create interactive user interfaces.
- Backend: HTML, PHP, JavaScript / PowerApps
- Dataset source: Kaggle, subject to change

Rational for Project Choice

I have chosen this project because it combines my interest in software development but also because I have made similar “health based” applications prior and I feel like I have a good understanding of these types of projects. With this I feel like I will be able to

connect with people and I am interested in exploring how an app can help users maintain motivation, recognise their progress and reflect on the reasons behind their goal(s).

My project is good for a final year project because it involves several areas of software development where I will create an interface, manage a database, full application development, data analysis and include testing and evaluation. It will allow me to apply my knowledge gained from previous university modules and further apply this to a practice problem and develop skills that are relevant to my future employment in software development / data analysis.

Another reason I have chose this project is the opportunity to combine application development with research rather than developing an application based only on assumptions.

The project also allows me to consider software design choices including how the app will look while also focusing on privacy, security and keeping a user's personal information safe.

Overall, I believe that while doing this project it will allow for me to create help solve a meaningful problem and evaluate how effectively the application meets its intended objectives

Background Research

The initial research will focus own understanding sobriety support applications; behaviour change and the role of apps that help people keep sober.

Existing applications provide a useful starting point for identifying common features and potential improvements. For example, there are already apps like I am Sober, Nomo and Try Dry which I can investigate to understand how they approach progress tracking, daily engagement and motivation.

The project will also investigate whether suitable datasets are available through Kaggle. It is important to recognise that publicly available datasets may not directly explain why people achieve or struggle with sobriety however if suitable data about motivations and recovery experiences cannot be found the analysis will focus on the questions that the available data can reasonably answer.

Areas for Investigation

Existing Apps

I will investigate exciting apps to compare their features, interface and approaches to supporting users. This will help identify common functionality, usability and potential opportunities for improvement.

Behaviour changes / motivation

I will explore psychological and behavioural concepts relevant to maintaining sobriety including goal setting, habit formation, self-reflection and motivation. This research will help inform the design of dial cheek ins, progress summaries and person reminders.

User experience and design

I will investigate how to create a clear, accessible and easy to navigate interface and make sure to pay particular attention to reduce unnecessary complexity to present clear progress and ensure the application uses supportive language.

Application development / database design

I will investigate technologies used to develop my app and I will consider how user information and daily check-ins can be stored and retrieved reliably.

Data Analysis and Dataset Suitability

I will investigate the availability of suitable datasets and evaluate their relevance, completeness and limitations.

Privacy, Security and Evaluation

I will investigate appropriate testing techniques to include

Research Ethics

Ethical consideration will be important because the project relates to sobriety, substance use sensitive personal experiences.

Firstly, any user research will be conducted only after obtaining the ethics approval. Participation will be voluntary and participants will receive clear information about the purpose of the research.

Secondly, the project will minimise the collection of sensitive personal information. Participants will not be asked to disclose details about substance use or recovery unless these are necessary for the research and have been approved prior.

Thirdly, any dataset will be checked and will be permitted to be used. Personal data will be handled in accordance with the data protection requirements including the UK GDPR and the Data Protection Act 2018.

Finally, I will follow my university's research ethics procedures through the project and seek guidance from my supervisor if any new ethical issues arise. No participant recruitment or collection of primary research data will begin until the required approval has been obtained.

Methodology (how)

I will be developing the project in stages, and these stages will be tested prior to launch and will be improved as the project progresses.

I will investigate existing apps, relevant research on motivation and behaviour change and the availability of appropriate datasets. I will define the target users, functional requirements and project objectives.

I will create wireframes and plan the user journey through the applications. I will also design the database structure, identify the required application components and decide on the development technologies.

Project Plan (what)

Phase	Task
Proposal and Planning	Finalise proposal, define scope and agree objectives with supervisor
Background Research	Review excising applications, academic research and potential datasets
Requirements and Design	Define requirements, create wireframes and design the database
Dataset Selection and Analysis	Identify a suitable dataset, clean data and conduct initial analysis
Core Development	Build the interface, authentication and sobriety tracking
Feature Development	Implement motivations, daily check ins, progress summaries and settings

Testing and Evaluation	Conduct functional testing and approved usability evaluation
Improvements	Resolve issues and refine the application based on findings
Final Report	Complete analysis, documentation and final report
Final Submission	Check requirements, prepare deliverable and submit the project

References

GitHub